New Uzbekistan University

Statement of Ethics: I agree to complete this exam without unauthorized assistance.

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Midterm Exam

COURSE NAME: Data Science Fundamentals COURSE NUMBER: CS203

EXAMINATION DURATION: 120 MINUTES EXAM VERSION: A

ADDITIONAL MATERIALS ALLOWED TO USE: Pens and a simple non-programmable calculator only.

Do not open this exam paper until you are told to begin.

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READ THESE INSTRUCTIONS FIRST. Breaking the rules can lead to a loss of points.

- Coverage: Weeks 1–5. Answer all 26 questions: 20 multiple-choice questions (MCQs) and 6 written questions in 5 problems. Total: 100 points. Use the spaces provided. If you continue on the back, write the question number there and note where your answer continues.
- Each question has its own data. Do not use data from another question. Each MCQ is worth 2 points. Wrong answers receive 0; no points are deducted. In written answers, show calculations and explain your reasons using the case given. Correct steps can earn points.
- Sign your name and write your student ID on every page.
- Follow the University Examination Rules. This is a closed-book exam. Do not use books, notes, laptops, phones, smartwatches, the internet or AI tools. Do not share calculators or other materials.

FOR INSTRUCTOR USE ONLY. STUDENTS MUST NOT WRITE IN THIS TABLE.

Problem	Type of Questions	Points	Score
1	Multiple Choice — Questions 1–20	40	
2	Short Answer — Questions 21–22	20	
3	Problem Solving — Questions 23–24	20	
4	Analytical Question — Question 25	10	
5	Open Question — Question 26	10	
Total		**100**	

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Problem 1 Multiple Choice (40 points)

Questions 1–5. Circle one answer (A–E) for each question. Each question is worth 2 points.

1. [2 points] A bike-share file has one row for each completed rental (one use of a bike by a rider). The same bike may appear in several rows. To find the mean duration of completed rentals, what is the unit of observation?

A. One distinct bike
B. One docking station
C. One calendar month
D. One completed rental
E. None of the above

2. [2 points] A survey asks riders about their satisfaction. It includes only riders who return bikes at East station (a bike collection and return point) before 09:00 on weekdays. Which change best improves coverage of all riders?

A. Interview many more riders at the same place and time
B. Ask the same riders more questions
C. Sample riders at different stations, times and days, using a written selection plan
D. Fill each unanswered question with the most common answer
E. None of the above

3. [2 points] A complete fleet list contains four bikes. Fleet means all the company's bikes. Yesterday, only one bike was rented. It was used for three separate trips. What fraction of the fleet was rented at least once?

A. 1/4
B. 3/4
C. 3/3
D. 1/3
E. None of the above

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4. [2 points] Four repairs take 2, 5, 5 and 8 minutes. Replace the last value with 80. Which statement is correct?

A. The median stays 5, but the mean increases
B. Both the median and mean stay 5
C. The median increases to 41.5
D. The mean stays 5, but the median increases
E. None of the above

5. [2 points] A clock error doubles every recorded rental duration. Compared with the correct durations, how do the recorded mean and variance change?

A. Both the mean and variance double
B. The mean stays the same and the variance doubles
C. The mean doubles and the variance is multiplied by four
D. The mean doubles and the variance stays the same
E. None of the above

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Problem 1 Multiple Choice continued (40 points)

Questions 6–10. Circle one answer (A–E) for each question. Each question is worth 2 points.

6. [2 points] Two records each contain repair durations for three bikes: A = [0, 10, 20] and B = [9, 10, 11] minutes. Both means are 10. Use sample variance with denominator n − 1. Which comparison is correct?

A. Their variances are equal because their means are equal
B. A has 10 times the variance of B
C. B has 100 times the variance of A
D. A has 100 times the variance of B
E. None of the above

7. [2 points] Every repair duration in a small record is 4 minutes. Their standard deviation (SD) is zero. What happens when you apply z = (x − mean)/SD to these values?

A. Every z-score automatically equals 1
B. The formula gives z = 4
C. A zero denominator proves that the original durations are invalid
D. The formula is undefined because the denominator is zero; a clear rule is needed to handle this case
E. None of the above

8. [2 points] Trip ID is a unique key: each Trip ID should identify one trip. Two rows have the same Trip ID but different return times and durations. What is the best first action?

A. Keep both rows as separate trips because their durations differ
B. Treat the rows as conflicting records and check the source before correcting them
C. Average their return times and use that as the true record
D. Delete the longer trip without checking
E. None of the above

9. [2 points] One rental duration is recorded as 2 hours. Other rows use minutes. The source confirms that 2 hours is correct. What should you use in an analysis measured in minutes?

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A. 2 minutes, because only the number matters
B. A missing value, because the duration is unusually large
C. 120 minutes, keeping a record of the original value and the conversion
D. The median duration, without checking the unit
E. None of the above

10. [2 points] In a bike inspection file, 0 means “checked and no fault found”. A blank means “not checked”. A script changes every blank to 0. What is the main problem?

A. It keeps the difference between unchecked bikes and bikes with no fault
B. It changes unknown results into no-fault results and can make the fault rate appear too low
C. It must increase the measured fault rate
D. It changes only the format, so it cannot affect any analysis
E. None of the above

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Problem 1 Multiple Choice continued (40 points)

Questions 11–15. Circle one answer (A–E) for each question. Each question is worth 2 points.

11. [2 points] Training records contain counts [0, 0, 100]. The second 0 is confirmed to be an accidental copy. Which pipeline gives an appropriate median for imputing other missing training counts?

A. Fit the median first, then remove the copy: keep 0
B. Delete both zeros, then fit the median: 100
C. Choose the median using the test records
D. Remove the confirmed copy, then fit the median: 50
E. None of the above

12. [2 points] A cycle count is the number of earlier rentals. Training cycle counts are 2, 4, missing and 6. The held-out test count is 100. A median imputer is fitted only on training data. Which value fills the training blank?

A. 5
B. 4
C. 28
D. 100
E. None of the above

13. [2 points] At 12:00, a model must predict whether a rental will end with a fault. Which feature is available at that time?

A. The number of earlier rentals, already stored at 11:59
B. The fault inspection result entered after the bike returns at 15:00
C. The final duration of this rental, which ends at 15:00
D. The repair code assigned at 16:00
E. None of the above

14. [2 points] The training minimum and maximum are 0 and 100. A separate test scaler uses 20 and 60. The same original value, 40, becomes 0.4 in training and 0.5 in testing. What is the best correction?

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A. Keep separate scalers because test values should define their own scale
B. Transform both sets with the fixed training scaler, giving 0.4 for 40
C. Refit the training scaler whenever a test value arrives
D. Replace every test value with the training mean
E. None of the above

15. [2 points] Five paired records give distance in km [1, 2, 3, 4, 5] and waiting time in minutes [4, 1, 0, 1, 4]. Values in the same position form a pair. Pearson correlation is 0. Which conclusion is justified?

A. A curved relationship may still be present; inspect the paired scatter plot
B. The variables have no relationship because the correlation is 0
C. Every waiting time must be equal because the correlation is 0
D. The result proves that distance causes the waiting time
E. None of the above

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Problem 1 Multiple Choice continued (40 points)

Questions 16–20. Circle one answer (A–E) for each question. Each question is worth 2 points.

16. [2 points] Rainy days have fewer rentals and more recorded faults than dry days. A plot that combines all days shows fewer rentals together with more faults. Which next step best checks whether weather explains this association?

A. Compare rental–fault patterns within each weather group and compare the makeup of those groups
B. Conclude that increasing rentals prevents faults
C. Delete every rainy day because it does not support the desired result
D. Ignore weather because a combined relationship always shows cause and effect
E. None of the above

17. [2 points] Three bikes aged 1, 2 and 3 years have 2, 4 and 6 recorded repairs. Adding a 20-year-old bike with only one recorded repair greatly changes the correlation. What is the best next step?

A. Delete the bike only because the correlation changed
B. Conclude that old bikes need fewer repairs
C. Check the paired plot, how much the bike was used, and where its record came from before interpreting or deleting the unusual point
D. Keep only the first three bikes because they form a straight line
E. None of the above

18. [2 points] A manager wants to check whether bike age and repair time have a linear or curved relationship. Both are numerical measurements recorded for each bike. Which display best keeps the information needed for this check?

A. A pie chart of the total number of bikes
B. Two separate histograms with no bike-level links between them
C. One bar showing only the mean repair time
D. A scatter plot with each bike's age and repair time as one point
E. None of the above

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19. [2 points] Station A has 1 faulty bike among 2 checked bikes. Station B has 20 among 40. At each station, one bike previously recorded as fault-free is found to have a fault. The numbers checked stay fixed. Which statement is correct?

A. Both fault percentages increase by the same amount because they started at 50%
B. A increases by 50 percentage points; B increases by 2.5 percentage points
C. B changes more because more bikes were checked there
D. Neither percentage changes because the numbers checked stay fixed
E. None of the above

20. [2 points] Bike A has 120 rental minutes over 6 trips. Bike B has 60 rental minutes over 3 trips. Both have a feature of 20 minutes per trip. What does using only this ratio hide?

A. Bike B has a longer average trip than Bike A
B. The two bikes must have the same total rental time
C. Their total activity differs; keep the totals and trip counts if the question needs that information
D. The ratio must increase whenever the number of trips increases
E. None of the above

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Problem 2 Short Answer (20 points)

Question 21 Counting bikes in service records [10 points]

A bike-share company rents bikes to riders. Its fleet (all its bikes) contains exactly the three bikes in the first table. The second table lists every service job from yesterday. Each Job ID is a different, genuine job. A bike may have several jobs or none.

Bike ID	Home station
B17	East
B24	West
B31	East

Job ID	Bike ID
J1	B17
J2	B17
J3	B24

(a) [3 points] State the unit of observation needed to find the fraction of fleet bikes serviced yesterday. Calculate that fraction.

(b) [4 points] A join keeps only bikes that have a matching service job. It gives three rows. A report says, “Every fleet bike was serviced because every joined row has a Job ID.”
Explain the error. Describe a join or counting method that keeps the correct denominator. Explain what happens to B17 and B31.

(c) [3 points] A separate rider survey includes only East station before 09:00 on weekdays.
Identify two groups of riders it may miss. Suggest one practical change to the sampling method so that a claim can cover all riders.

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Problem 2 Short Answer continued (20 points)

Question 22 Comparing two repair records [10 points]

The table gives two separate repair records, each covering all five bikes in its group. Downtime is the time a bike cannot be used. The values show downtime in minutes per bike. Row numbers only list the values: East and West values in the same row are not a matched pair. Zero downtime is valid when a bike has no interruption.

Record row	East minutes	West minutes
1	2	3
2	2	3
3	4	4
4	4	5
5	8	5

(a) [3 points] Calculate the mean, median and range of the East values.

(b) [4 points] The West values also have mean and median 4. A report says this proves that the two groups have the same downtime behaviour.
Use specific table values to assess this claim. Which group has more consistent downtime in this sample? What property is not described by the mean and median?

(c) [3 points] Add one valid East observation of 0 minutes. Do not recalculate every summary. Predict whether the East mean and range each rise, fall or stay the same. Explain both predictions and give the new range.

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Problem 3 Problem Solving (20 points)

Question 23 Cleaning rental data with different units [10 points]

Each Trip ID should identify one rental (one use of a bike). The data source confirms three facts: the second T02 row is an accidental extra copy; −4 is an invalid duration; and T05 really lasted four hours. A blank or invalid duration means unknown, not zero. Use minutes for the analysis.

Trip ID	Bike ID	Duration	Unit
T01	B17	12	minutes
T02	B24	0.5	hours
T02	B24	0.5	hours
T03	B31	15	minutes
T04	B40	−4	minutes
T05	B17	240	minutes

(a) [4 points] State what you would do with each of these: the repeated T02 row, 0.5 hours, the invalid −4, and the confirmed 240 minutes.
Explain why a large value alone is not a reason to remove the last value.

(b) [3 points] After these actions, the mean of the four valid recorded durations is 74.25 minutes. A report calls this “the mean duration of all five rentals”.
Is this description justified? Explain the problem caused by the missing information. Give an accurate description of this mean.

(c) [3 points] For a separate summary, fill the one unknown duration with the median of the four valid durations.
Calculate the fill value. State two facts that should remain clear when you report this imputation.

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Problem 3 Problem Solving continued (20 points)

Question 24 Checking preprocessing before bikes leave [10 points]

At morning dispatch (when bikes are sent out), a model predicts whether a bike will return with a fault. Cycle count is the number of earlier rentals already recorded by dispatch. The home station is also known at dispatch. Station is a nominal category with no ranking. Keep the training/test split in the table. Use only the given observed counts for median imputation.

Bike	Data split	Cycle count	Station
U1	Training	10	North
U2	Training	Missing	South
U3	Training	30	East
U4	Training	50	North
U5	Test	90	East

(a) [4 points] A colleague fits a median imputer using all observed rows before applying the fixed split. Give the fill value from all observed rows and the fill value from training rows only. Which is appropriate for held-out evaluation? Explain why using no test labels does not answer the concern.

(b) [3 points] The colleague codes North = 1, South = 2 and East = 3, then uses numerical distances between stations.
Explain the assumption that is not supported by these labels. Suggest another encoding. State what should determine whether two station labels are treated as different.

(c) [3 points] The colleague uses “fault confirmed after return” as an input because it is strongly related to the target.
Explain the error. Name one given feature that can be used at dispatch. Explain when that feature must be available.

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Problem 4 Analytical Question (10 points)

Question 25 What separate data lists can tell you [10 points]

A station-day means one station on one day. For four station-days, rental counts and total repair minutes were saved in the two separate sorted lists below. Their links to dates were lost. Values in the same list position are NOT known to be from the same day. Use every rental count and every repair-minute value once.

File 1 Rental counts	Matching dates **are missing**	File 2 Repair minutes
10, 20, 30, 40		20, 40, 60, 80

(a) [4 points] Create one possible set of four daily pairs with a perfectly positive linear relationship (an upward straight line). Create another set with a perfectly negative linear relationship (a downward straight line).
List the pairs or draw two clearly labelled sketches.

(b) [3 points] A report says, “The two lists have fixed means and histograms, so they also determine the direction of the daily correlation.”
Use your examples to assess this claim. State what information was lost and what data would solve the problem.

(c) [3 points] Suppose the correct dates are found and the daily correlation is positive. Does this show that more rentals cause more repair time?
Give one other explanation involving a third variable. Suggest one further comparison or measurement that would help check your explanation.

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Problem 5 Open Question (10 points)

Question 26 Compare stations before and after combining groups [10 points]

These bike-inspection records come from two stations. Bikes were not randomly assigned to stations. “Use group” means how much a bike was ridden before inspection. Each bike is in one group and was checked once, using the same fault checklist. Each cell shows bikes with a fault / bikes checked.

Use group	East: faults / checked	West: faults / checked
Low-use	2 / 20	6 / 40
High-use	18 / 30	7 / 10

(a) [3 points] Calculate the fault percentage at each station within each use group.
Which station has the lower percentage in both groups?

(b) [4 points] A report combines the use groups and calls East less reliable because its overall fault percentage is higher.
Calculate the two overall percentages. Explain why this overall comparison gives a different ranking from part (a), using the numbers in each use group.

(c) [3 points] Suppose each station had equal numbers of low-use and high-use bikes, and its fault percentage in each group stayed the same. Which station would then have the lower overall percentage? Explain.
Do these records prove that the station itself causes the difference? Explain.

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END OF EXAM

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